

Centered or Close?

Unit 3 · Topic 3.1 · THE PROBLEM

Suppose a research team tests two rules for estimating a support proportion in a validation population where the true proportion is $p = 0.40$. For a random sample of 25 people, let X be the number who support the proposal.

Rule A reports $\hat{p}_A = X/25$. Rule B reports $\hat{p}_B = (X + 5)/35$. Exact summaries of the two sampling distributions are shown.

Lena: “A is unbiased, so it is always the more trustworthy estimator.”

Omar: “B varies less and is more likely to land within 0.06 of the truth, so its bias does not matter.”

Sampling distributions when $p = 0.40$

Rule	Mean	SD	Within 0.06
A	0.400	0.098	0.459
B	0.429	0.070	0.659

FIRST TAKE — one sentence before you work the parts

Which rule looks more useful for getting close to 0.40? Use one entry in the table.

- DOK 1** (a) If one sample has $X = 9$ supporters, calculate the point estimate from Rule A and from Rule B. State the population parameter both rules estimate.
- DOK 2** (b) Use the table to classify each rule as biased or unbiased. For any biased rule, give the direction and amount of bias; then compare the rules' sampling variability.
- DOK 3** (c) In 3–4 sentences, choose a rule for getting within 0.06 of p and use the table to defend it. Explain what Lena and Omar each overlook about bias and variability.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.